NEWTON SCHOOL · ANSWER KEY

PRE-CALCULUS | 🔑 ANSWER KEY

Recursion: Tower of Hanoi & Characteristic Equation

Student: Justin Koo **Date:** 2026-04-26 **Pages:** 2 problems · full solutions

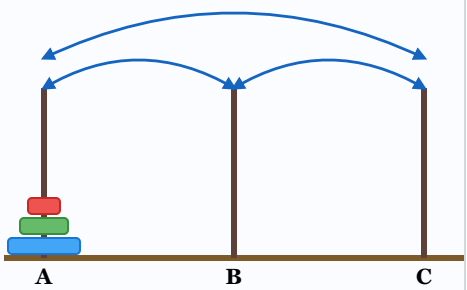
Standards: *CCSS.MATH.CONTENT.HSF-IF.A.3 · HSF-BF.A.2*

Problem 1 · Tower of Hanoi — Variants & Shift Trick

📘 Setup — Why the recurrences differ

Original Hanoi

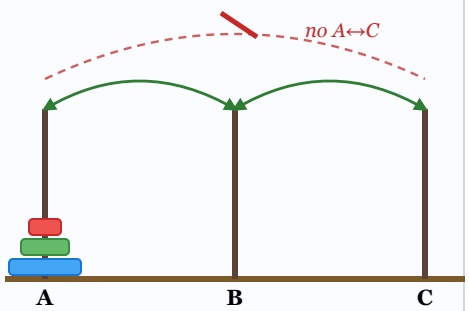
*disks may move **between any two pegs***



$$T_n = 2T_{n-1} + 1, \quad T_1 = 1$$

Adjacent-Only Hanoi

*disks move only **between adjacent pegs** (no A↔C)*



$$A_n = 3A_{n-1} + 2, \quad A_1 = 2$$

Key idea. In *original* Hanoi, the largest disk moves once (A → C directly). In *adjacent-only*, it must hop A → B → C (two moves). Between the two big-disk hops, the smaller stack of $n-1$ disks has to be moved **three** times: A→C, then C→A, then A→C — hence $3A_{n-1} + 2$ instead of $2T_{n-1} + 1$.

📏 SOLVED EXAMPLE A *Original Tower of Hanoi — closed form via shift trick*

Solve $T_n = 2T_{n-1} + 1$ with $T_1 = 1$. (This is the ORIGINAL Hanoi recurrence — every textbook starts here.)

Step 1. Substitute $U_n = T_n + c$, so $T_n = U_n - c$:

$$U_n - c = 2(U_{n-1} - c) + 1 \;\Rightarrow\; U_n = 2U_{n-1} + (1 - c)$$

Kill the constant: $1 - c = 0 \;\Rightarrow\; \boxed{c = 1}$.

Fixed-point check: $T^* = 2T^* + 1 \Rightarrow T^* = -1$, so $c = -T^* = 1$ ✓.

Step 2. $U_n = 2U_{n-1}$ with $U_1 = T_1 + 1 = 1 + 1 = 2$:

$$U_n = U_1 \cdot 2^{n-1} = 2 \cdot 2^{n-1} = 2^n$$

Step 3. Un-substitute: $T_n = U_n - 1 = 2^n - 1$.

Step 4 (verify). $T_2 = 2(1) + 1 = 3$, closed: $2^2 - 1 = 3$ ✓ · $T_3 = 2(3) + 1 = 7$, closed: $2^3 - 1 = 7$ ✓

$$\boxed{T_n = 2^n - 1}$$

SOLVED EXAMPLE B ***Another shift-trick example — different coefficients***

Solve $S_n = 2S_{n-1} + 5$ with $S_1 = 3$.

Step 1. Substitute $U_n = S_n + c$:

$U_n - c = 2(U_{n-1} - c) + 5 \;\Rightarrow\; U_n = 2U_{n-1} + (5 - c)$

Kill the constant: $5 - c = 0 \;\Rightarrow\; \boxed{c = 5}$.

Fixed-point check: $S^* = 2S^* + 5 \Rightarrow S^* = -5$, so $c = -S^* = 5$ ✓.

Step 2. $U_n = 2U_{n-1}$ with $U_1 = S_1 + 5 = 3 + 5 = 8$:

$U_n = 8 \cdot 2^{n-1} = 2^3 \cdot 2^{n-1} = 2^{n+2}$

Step 3. Un-substitute: $S_n = U_n - 5 = 2^{n+2} - 5$.

Step 4 (verify).

Recurrence: $S_2 = 2(3) + 5 = 11$. Closed: $2^4 - 5 = 16 - 5 = 11$ ✓

Recurrence: $S_3 = 2(11) + 5 = 27$. Closed: $2^5 - 5 = 32 - 5 = 27$ ✓

$$\boxed{S_n = 2^{n+2} - 5}$$

SOLVED EXAMPLE C ***Medication dosage — same recurrence, two ways to solve***

Setup. A patient takes the same pill every day. The body metabolizes 40% of the drug daily (so 60% remains), and the patient adds a 20 mg dose. Let M_n = drug level in the body at the start of day n.

$$M_n = 0.6\, M_{n-1} + 20, \quad M_0 = 0 \text{ mg}$$

This is the same $a_n = r\, a_{n-1} + b$ shape as Hanoi — but with $|r| < 1$, so the sequence ***converges*** *instead of growing without bound. That opens up a second, faster method.*

Method 1 — Shift Trick (+c substitution)

Step 1. Substitute $U_n = M_n - L$, treating L as a constant chosen to kill the "+20":

$U_n + L = 0.6(U_{n-1} + L) + 20 \;\Rightarrow\; U_n = 0.6\, U_{n-1} + (20 - 0.4L)$

Set $20 - 0.4L = 0 \;\Rightarrow\; \boxed{L = 50}$.

Step 2. $U_n = 0.6\, U_{n-1}$ with $U_0 = M_0 - L = 0 - 50 = -50$:

$U_n = -50 \cdot (0.6)^n$

Step 3. Un-substitute: $M_n = U_n + L = 50 - 50(0.6)^n = 50\big(1 - (0.6)^n\big)$.

$$\boxed{M_n = 50\big(1 - (0.6)^n\big)}$$

Method 2 — Limit Method ($M_n = M_{n-1}$ as $n \to \infty$)

Idea: since $|r| = 0.6 < 1$, the sequence converges. Call the limit $L = \lim_{n\to\infty} M_n$. Then M_{n-1} also approaches L.

Step 1. Take the limit of both sides of the recurrence:

$$\lim_{n\to\infty} M_n \;=\; 0.6 \cdot \lim_{n\to\infty} M_{n-1} + 20$$

$$L \;=\; 0.6\,L + 20 \;\Rightarrow\; 0.4\,L = 20 \;\Rightarrow\; \boxed{L = 50 \text{ mg}}$$

Pharmacology calls this the ***steady-state concentration****.*

Step 2. Apply the standard closed-form formula for first-order linear recurrences (whenever the limit exists):

$$M_n = L + (M_0 - L)\,r^n$$

Why this works: the "distance from the limit" $D_n = M_n - L$ satisfies $D_n = r\,D_{n-1}$ — pure geometric — so $D_n = D_0 \cdot r^n$. Add L back to get M_n.

Step 3. Plug in $L = 50, M_0 = 0, r = 0.6$:

$$M_n = 50 + (0 - 50)(0.6)^n = 50 - 50(0.6)^n = 50\big(1 - (0.6)^n\big)$$

$$\boxed{M_n = 50\big(1 - (0.6)^n\big)} \quad \textbf{— same answer ✓}$$

Behavior — convergence to the limit:

Day n	M_n (mg)	Distance from $L = 50$
0	0	50
1	20	30
2	32	18
5	46.11	3.89
10	49.70	0.30
∞	50	0

The distance from L shrinks by a factor of 0.6 each day — that's exactly $D_n = 50 \cdot (0.6)^n$. After 10 days you're within 0.30 mg of the steady state.

When to use which method:

- **Shift trick** works *always* — Hanoi ($|r| = 2 > 1$, no limit), medication ($|r| = 0.6 < 1$, has limit), anything in between.

- **Limit method** only works when $|r| < 1$ (sequence actually converges) — but when it does, it's a one-line argument plus a plug-in.
- Both lead to the same closed form. They're *algebraically identical*: $L = -c$ in our shift-trick notation, since L is the fixed point and $c = -L$ is what kills the constant.

(1a) Sketch a 3-disk move count. (1b) Solve $A_n = 3A_{n-1} + 2, A_1 = 2$ via shift trick. (1c) Repeat the shift trick on $Q_n = 4Q_{n-1} + 3, Q_1 = 5$.

1a. Move counts for $n = 3$.

$T_3 = 2T_2 + 1 = 2(\boxed{3}) + 1 = \boxed{7}$

$A_3 = 3A_2 + 2 = 3(\boxed{8}) + 2 = \boxed{26}$

Adjacent Hanoi takes more than 3× as many moves — not surprising, since the constraint is tighter.

1b. Shift trick for $A_n = 3A_{n-1} + 2,\ A_1 = 2$.

Step 1. Substitute $U_n = A_n + c$, so $A_n = U_n - c$:

$U_n - c = 3(U_{n-1} - c) + 2 \ \Rightarrow\ U_n = 3U_{n-1} - 3c + c + 2 = 3U_{n-1} + (2 - 2c)$

For pure geometric: $2 - 2c = 0 \ \Rightarrow\ \boxed{c = 1}$.

Step 2. $U_n = 3U_{n-1}$ with $U_1 = A_1 + 1 = 2 + 1 = 3$:

$U_n = U_1 \cdot 3^{n-1} = 3 \cdot 3^{n-1} = 3^n$

Step 3. Un-substitute: $A_n = U_n - 1 = 3^n - 1$.

Step 4. Verify:

A_2: recurrence gives $3(2) + 2 = 8$; closed form gives $3^2 - 1 = 8$ ✓

A_3: recurrence gives $3(8) + 2 = 26$; closed form gives $3^3 - 1 = 26$ ✓

PROBLEM 1b — FINAL ANSWER

$$\boxed{A_n = 3^n - 1}$$

Why the shift works: *the fixed point of the map $A \mapsto 3A + 2$ is $A^* = -1$ (since $-1 = 3(-1) + 2$). Shifting by $c = -A^* = 1$ kills the constant — the map then sends $0 \to 0$ in U-coordinates, i.e. pure geometric.*

1c. New recurrence: $Q_n = 4Q_{n-1} + 3,\ Q_1 = 5$.

Step 1. Substitute $U_n = Q_n + c$, so $Q_n = U_n - c$:

$U_n - c = 4(U_{n-1} - c) + 3 \;\Rightarrow\; U_n = 4U_{n-1} + (3 - 3c)$

Kill the constant: $3 - 3c = 0 \;\Rightarrow\; \boxed{c = 1}$.

Quick check via fixed point: $Q^* = 4Q^* + 3 \Rightarrow Q^* = -1$, so $c = -Q^* = 1$ ✓.

Step 2. $U_n = 4U_{n-1}$ with $U_1 = Q_1 + 1 = 5 + 1 = 6$:

$U_n = U_1 \cdot 4^{n-1} = 6 \cdot 4^{n-1} = \frac{6}{4} \cdot 4^n = \frac{3}{2} \cdot 4^n$

Step 3. Un-substitute: $Q_n = U_n - 1 = \frac{3}{2} \cdot 4^n - 1$, or equivalently $Q_n = 6 \cdot 4^{n-1} - 1$.

Step 4. Verify:

Q_2 from recurrence: $4(5) + 3 = 23$.

Q_2 from closed form: $6 \cdot 4^{2-1} - 1 = 6 \cdot 4 - 1 = 24 - 1 = 23$ ✓

Q_3 recurrence: $4(23) + 3 = 95$. Closed form: $6 \cdot 4^2 - 1 = 96 - 1 = 95$ ✓

PROBLEM 1c — FINAL ANSWER

$$\boxed{Q_n = 6 \cdot 4^{n-1} - 1 \;=\; \frac{3}{2} \cdot 4^n - 1}$$

Problem 2 · Characteristic Equation Method

THEOREM ***Characteristic Equation for 2nd-Order Linear Homogeneous Recurrence***

For $a_n = p \cdot a_{n-1} + q \cdot a_{n-2}$ with given a_0, a_1:

1. Write the **characteristic equation**: $x^2 - p\,x - q = 0$.
2. Solve for roots r_1, r_2.
3. **Distinct roots:** $a_n = C_1\, r_1^n + C_2\, r_2^n$.
4. Plug in $a_0, a_1 \to$ solve the 2×2 system for C_1, C_2.
5. Substitute back. Closed form gives a_n for any n without the recurrence.

Three Cases of Roots — Solved Examples

The discriminant $\Delta = p^2 + 4q$ of $x^2 - px - q = 0$ tells you which case you're in. The general-solution *shape* changes — but Steps 1, 4, 5 of the theorem are the same.

CASE 1 ***Two distinct (integer) roots — general solution:*** $a_n = C_1 r_1^n + C_2 r_2^n$

Solve $a_n = 3a_{n-1} - 2a_{n-2}$ with $a_0 = 1,\ a_1 = 3$.

Step 1. $p = 3, q = -2$. Char eqn: $x^2 - 3x + 2 = 0$.

Step 2. Factor: $(x-1)(x-2) = 0 \ \Rightarrow\ r_1 = 1,\ r_2 = 2$. ($\Delta = 9 - 8 = 1 > 0$, distinct.)

Step 3. General: $a_n = C_1 \cdot 1^n + C_2 \cdot 2^n = C_1 + C_2 \cdot 2^n$.

Step 4. Use initials:

$a_0 = 1$: $C_1 + C_2 = 1$ ...(i)

$a_1 = 3$: $C_1 + 2C_2 = 3$ ...(ii)

(ii) − (i): $C_2 = 2$, so $C_1 = -1$.

Step 5. Closed form: $a_n = -1 + 2 \cdot 2^n = 2^{n+1} - 1$.

$$\boxed{a_n = 2^{n+1} - 1}$$

Check: a_2 recurrence $= 3(3) - 2(1) = 7$; closed $= 2^3 - 1 = 7$ ✓

CASE 2 ***Repeated root r — general solution:*** $a_n = (C_1 + C_2\, n) r^n$

Solve $b_n = 4b_{n-1} - 4b_{n-2}$ with $b_0 = 1,\ b_1 = 4$.

Step 1. $p = 4, q = -4$. Char eqn: $x^2 - 4x + 4 = 0$.

Step 2. Factor: $(x-2)^2 = 0 \ \Rightarrow\ r = 2$ (double root). ($\Delta = 16 - 16 = 0$.)

Step 3. General (repeated-root form): $b_n = (C_1 + C_2\, n) \cdot 2^n$.

Why the extra n? If we used $C_1 r^n + C_2 r^n = (C_1 + C_2) r^n$, the two terms collapse — only one constant of freedom. The $n \cdot r^n$ term is the second linearly-independent solution.

Step 4. Use initials:

$b_0 = 1$: $(C_1 + 0) \cdot 2^0 = C_1 = 1$.

$b_1 = 4$: $(1 + C_2) \cdot 2 = 4 \;\Rightarrow\; C_2 = 1$.

Step 5. Closed form: $b_n = (1 + n) \cdot 2^n = (n + 1)\, 2^n$.

$$\boxed{b_n = (n+1)\, 2^n}$$

Check: b_2 recurrence $= 4(4) - 4(1) = 12$; closed $= (3)(4) = 12$ ✓ | b_3 recurrence $= 4(12) - 4(4) = 32$; closed $= (4)(8) = 32$ ✓

CASE 3 ***Complex roots $\alpha \pm \beta i$ — general solution:***
$a_n = C_1(\alpha + \beta i)^n + C_2(\alpha - \beta i)^n$

Note: we keep complex powers as-is — no Euler / no $\cos\theta$, $\sin\theta$. Conjugate pairs guarantee the result is real.

Solve $c_n = 2c_{n-1} - 2c_{n-2}$ with $c_0 = 0,\ c_1 = 1$.

Step 1. $p = 2, q = -2$. Char eqn: $x^2 - 2x + 2 = 0$.

Step 2. Quadratic formula: $x = \dfrac{2 \pm \sqrt{4-8}}{2} = \dfrac{2 \pm \sqrt{-4}}{2} = \dfrac{2 \pm 2i}{2} = 1 \pm i$. ($\Delta = 4 - 8 = -4 < 0$, complex.)

Roots: $r_1 = 1 + i,\quad r_2 = 1 - i$.

Step 3. General (complex number form, conjugates):

$c_n = C_1(1+i)^n + C_2(1-i)^n$

Step 4. Use initials:

$c_0 = 0$: $C_1 + C_2 = 0 \;\Rightarrow\; C_2 = -C_1$ ...(i)

$c_1 = 1$: $C_1(1+i) + C_2(1-i) = 1$. Substitute (i):

$C_1(1+i) - C_1(1-i) = 1 \;\Rightarrow\; C_1\big[(1+i) - (1-i)\big] = C_1(2i) = 1$

$C_1 = \dfrac{1}{2i} = \dfrac{1}{2i} \cdot \dfrac{-i}{-i} = \dfrac{-i}{-2i^2} = \dfrac{-i}{2} = -\dfrac{i}{2}$, and $C_2 = \dfrac{i}{2}$.

Step 5. Closed form (factor out $\frac{i}{2}$):

$$c_n = -\frac{i}{2}(1+i)^n + \frac{i}{2}(1-i)^n = \frac{i}{2}\Big[(1-i)^n - (1+i)^n\Big]$$

$$\boxed{c_n = \frac{i}{2}\big[(1-i)^n - (1+i)^n\big]}$$

Why the answer is REAL (always, for complex-conjugate roots):

Since $C_2 = \overline{C_1}$ and $(1-i)^n = \overline{(1+i)^n}$, the two terms are conjugates. *Conjugate plus conjugate is twice the real part — i.e. real.* So the imaginary parts cancel and c_n is always a real integer (in this example).

Verification for small n:

c_2 from recurrence: $2(1) - 2(0) = 2$.

c_2 from closed form: $(1+i)^2 = 1 + 2i + i^2 = 2i$; $(1-i)^2 = -2i$;

$c_2 = \frac{i}{2}\left[(-2i) - (2i)\right] = \frac{i}{2}(-4i) = -2i^2 = 2$ ✓

c_3 from recurrence: $2(2) - 2(1) = 2$.

c_3 from closed form: $(1+i)^3 = (1+i)(2i) = 2i + 2i^2 = -2 + 2i$; $(1-i)^3 = -2 - 2i$;

$c_3 = \frac{i}{2}\left[(-2-2i) - (-2+2i)\right] = \frac{i}{2}(-4i) = 2$ ✓

Sequence preview: $c_n = 0, 1, 2, 2, 0, -4, -8, -8, 0, 16, \ldots$ — periodic in sign because the modulus $|1+i| = \sqrt{2}$ grows while the argument rotates by 45° each step.

Summary table:

Discriminant $\Delta = p^2 + 4q$	Roots	General solution
$\Delta > 0$ (Case 1)	$r_1 \neq r_2$ real	$a_n = C_1 r_1^n + C_2 r_2^n$
$\Delta = 0$ (Case 2)	$r_1 = r_2 = r$	$a_n = (C_1 + C_2 n) r^n$
$\Delta < 0$ (Case 3)	$\alpha \pm \beta i$ conjugate pair	$a_n = C_1(\alpha + \beta i)^n + C_2(\alpha - \beta i)^n$

(2a) Apply the theorem to $L_n = L_{n-1} + L_{n-2}$, $L_0 = L_1 = 1$. Show all three forms of the general solution.

Step 1. Identify $p = 1, q = 1$. Characteristic equation: $x^2 - x - 1 = 0$.

Step 2. Quadratic formula:

$$x = \frac{1 \pm \sqrt{1+4}}{2} = \frac{1 \pm \sqrt{5}}{2}$$

Name the roots:

$$\varphi = \frac{1+\sqrt{5}}{2} \approx 1.6180, \quad \psi = \frac{1-\sqrt{5}}{2} \approx -0.6180$$

Step 3. General solution — three equivalent forms:

Symbolic form: $L_n = C_1\,\varphi^n + C_2\,\psi^n$

Radical form: $L_n = C_1\left(\frac{1+\sqrt{5}}{2}\right)^n + C_2\left(\frac{1-\sqrt{5}}{2}\right)^n$

Decimal form: $L_n \approx C_1(1.6180)^n + C_2(-0.6180)^n$

Step 4. Use $L_0 = L_1 = 1$:

$L_0 = 1$: $C_1 + C_2 = 1$ ...(i)

$L_1 = 1$: $C_1\varphi + C_2\psi = 1$ ...(ii)

From (i): $C_2 = 1 - C_1$. Substitute into (ii):

$C_1\varphi + (1 - C_1)\psi = 1 \;\Rightarrow\; C_1(\varphi - \psi) = 1 - \psi$

Compute $\varphi - \psi = \sqrt{5}$ and $1 - \psi = 1 - \frac{1-\sqrt{5}}{2} = \frac{1+\sqrt{5}}{2} = \varphi$. Hence

$$C_1 = \frac{\varphi}{\sqrt{5}}, \quad C_2 = 1 - C_1 = \frac{-\psi}{\sqrt{5}}$$

Step 5. Closed form (collect):

$$L_n = \frac{\varphi \cdot \varphi^n - \psi \cdot \psi^n}{\sqrt{5}} = \frac{\varphi^{n+1} - \psi^{n+1}}{\sqrt{5}}$$

This is Binet's formula shifted by 1: $L_n = F_{n+1}$, where $F_k = (\varphi^k - \psi^k)/\sqrt{5}$ is the classical Fibonacci.

PROBLEM 2a — FINAL ANSWER

$$\boxed{L_n = \frac{\varphi^{n+1} - \psi^{n+1}}{\sqrt{5}}, \quad \varphi = \tfrac{1+\sqrt{5}}{2}, \quad \psi = \tfrac{1-\sqrt{5}}{2}}$$

Sanity check via decimal form: compute L_5.

$\varphi^6 \approx (1.6180)^6 \approx 17.944$

$\psi^6 \approx (-0.6180)^6 \approx 0.0557$

$L_5 \approx \dfrac{17.944 - 0.0557}{\sqrt{5}} \approx \dfrac{17.888}{2.2361} \approx 8.000$ ✓

Recurrence build-up: $1, 1, 2, 3, 5, 8 \;\Rightarrow\; L_5 = 8$. Match!

(2b) Apply the theorem to $a_n = 5a_{n-1} - 6a_{n-2}$, $a_0 = 0, a_1 = 1$.

Step 1. $p = 5, q = -6$. Characteristic equation:

$$x^2 - 5x + 6 = 0$$

Step 2. Factor: $(x-2)(x-3) = 0$, so $r_1 = 2,\ r_2 = 3$.

Step 3. General solution:

$$a_n = C_1 \cdot 2^n + C_2 \cdot 3^n$$

Step 4. Use $a_0 = 0, a_1 = 1$:

$a_0 = 0$: $C_1 + C_2 = 0$ ...(i)

$a_1 = 1$: $2C_1 + 3C_2 = 1$ ...(ii)

From (i): $C_1 = -C_2$. Substitute into (ii): $-2C_2 + 3C_2 = 1 \Rightarrow C_2 = 1,\ C_1 = -1$.

Step 5. Closed form: $a_n = -2^n + 3^n = 3^n - 2^n$.

PROBLEM 2b — FINAL ANSWER

$$\boxed{a_n = 3^n - 2^n}$$

Verify:

From recurrence ($a_0 = 0, a_1 = 1, a_n = 5a_{n-1} - 6a_{n-2}$):

$a_2 = 5(1) - 6(0) = 5$

$a_3 = 5(5) - 6(1) = 25 - 6 = 19$

$a_4 = 5(19) - 6(5) = 95 - 30 = 65$

From closed form $a_n = 3^n - 2^n$:

$a_2 = 9 - 4 = 5$ ✓ | $a_3 = 27 - 8 = 19$ ✓ | $a_4 = 81 - 16 = 65$ ✓

✅ Boxed Answers

- **P1a:** $T_3 = 7,\ A_3 = 26$
- **P1b:** $A_n = 3^n - 1$
- **P1c:** $Q_n = 6 \cdot 4^{n-1} - 1 = \frac{3}{2} \cdot 4^n - 1$
- **P2a:** $L_n = (\varphi^{n+1} - \psi^{n+1})/\sqrt{5}$
- **P2b:** $a_n = 3^n - 2^n$

🧭 Method recap

Recurrence type	Method	Trick
$a_n = r \cdot a_{n-1} + b$ *(1st-order, non-homogeneous)*	Shift trick	Find fixed point $a^* = b/(1-r)$, set $c = -a^*$, solve geometric.
$a_n = p \cdot a_{n-1} + q \cdot a_{n-2}$ *(2nd-order, homogeneous)*	Characteristic equation	Solve $x^2 - px - q = 0$. Use roots in $a_n = C_1 r_1^n + C_2 r_2^n$, fit C_1, C_2 to initial values.

Created by Newton School 2026-04-26 · Answer Key © Copyright